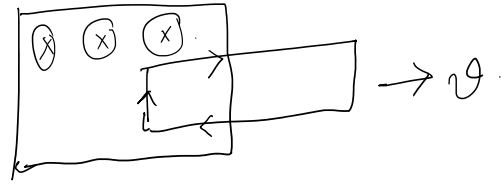


11-09-24, FO - Faraday's law. 1813

11 September 2024 08:02

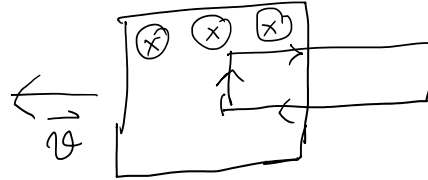
Expt. 1. Motional emf.

$$\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt}$$



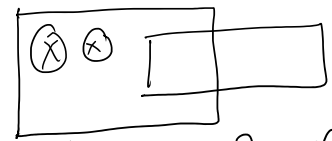
Magnetic flux $\phi = \int \mathbf{B} \cdot d\mathbf{a}$

Expt. 2. loop static $\rightarrow$ changes are static.



$\Rightarrow$ A changing magnetic field inducing electric field.

Exp. 3



changing B over time

$$\begin{aligned} \mathcal{E} &= \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\phi}{dt} \\ &= \oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a} = -\frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \end{aligned}$$

$$\Rightarrow \boxed{\oint \mathbf{E} \cdot d\mathbf{l} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a}}$$

Integral form of Faraday's law.

Stoke's law, $\oint \mathbf{E} \cdot d\mathbf{l} = \int \nabla \times \mathbf{E} \cdot d\mathbf{a}$

$$\Rightarrow \boxed{\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}}$$

Differential form of Faraday's law

For static magnetic fields, $\frac{\partial \mathbf{B}}{\partial t} = 0$

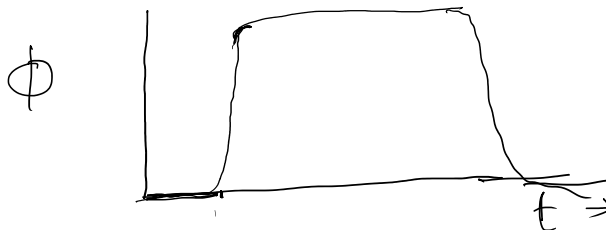
For static magnetic fields, $\frac{\partial B}{\partial t} = 0$

$$\Rightarrow \nabla \times E = \oint E \cdot dl = 0.$$

Lenz's law

$$E = - \frac{d\phi}{dt}$$

Ex. 7.5



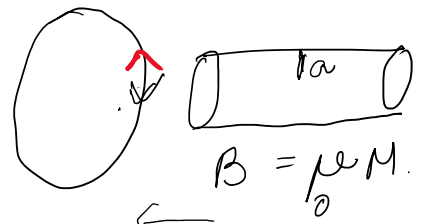
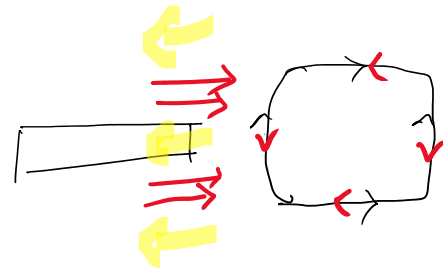
$$\phi = \int B \cdot da = \pi a^2 \mu_0 M.$$



Implication of Faraday.

$$\nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$



$$E = - \frac{\partial \phi}{\partial t}$$

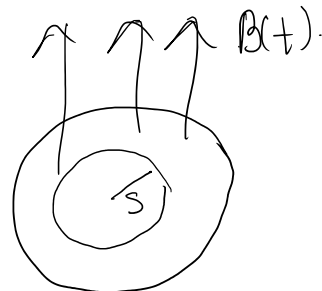
a ? black / red
b ? black / red

Ex. 7.7

$$\oint E \cdot dl = - \frac{d\phi}{dt}$$

$$E 2\pi S = - \frac{d}{dt} B(t) \pi S^2$$

$$\Rightarrow E = - \frac{S}{2} \frac{dB}{dt} \hat{\phi}$$



$$\mu_0 I / 2\pi S.$$

Ex. 7.9

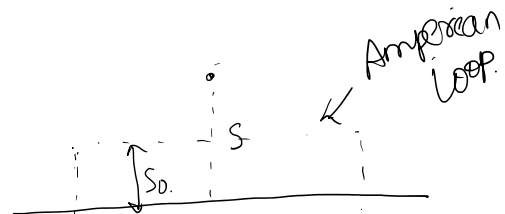
Amperean
loop.

Ex 7.9

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d}{dt} \int \vec{B} \cdot d\vec{a}$$

$$\vec{E}(s) = \left[\frac{\mu_0}{2\pi} \frac{dI}{dt} \ln s + k \right] \hat{z}$$

$\xrightarrow{L}$
 $\xrightarrow{I}$
 const. & int.



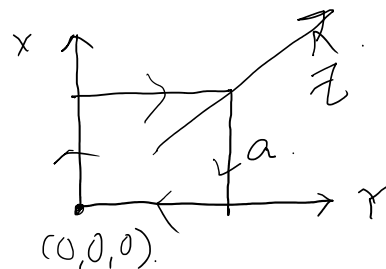
Prob. 7.13

A square loop of wire is in xy -plane with sides a , corner is at the origin.

non uniform time dependent mag. field

$$\vec{B}(t) = k y^3 t^2 \hat{z}$$

What is the electric field induced?



$$\phi = \int \vec{B} \cdot d\vec{a}$$

$$d\vec{a} = dx dy \hat{z}$$

$$d\vec{a} = \vec{dy} \times \vec{dy}$$

$$\phi = \int_0^a \int_0^a k y^3 t^2 (dx dy \hat{z}) (\hat{z})$$

$$= \int_0^a dx \int_0^a dy k y^3 t^2$$

$$= \frac{k t^2 a^5}{4}$$

$$\vec{E} = - \frac{d\phi}{dt} = - \frac{k a^5 t}{2}$$